

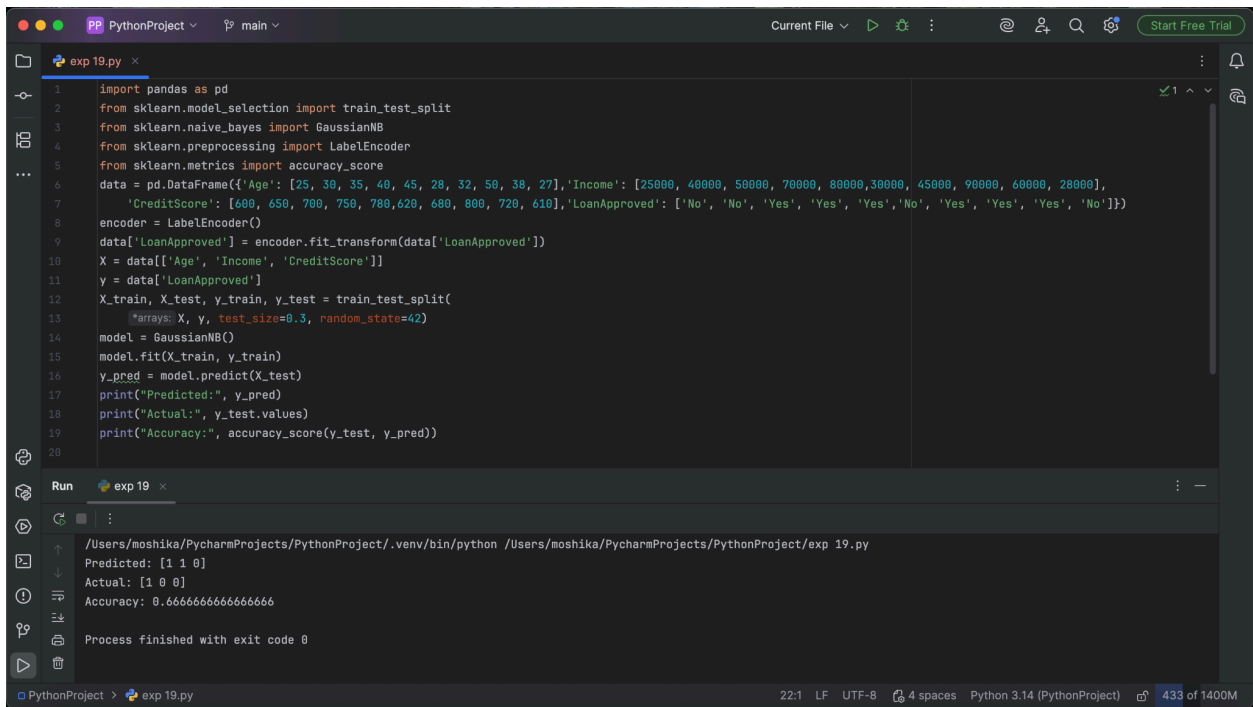
EXPERIMENT - 19

Aim: To predict whether a customer is eligible for a bank loan using the Naive Bayes classifier.

Procedure:

1. Load the bank loan dataset.
2. Preprocess the data and handle missing values.
3. Select relevant customer attributes.
4. Separate input features and loan approval status.
5. Split the data into training and testing sets.
6. Create a Naive Bayes classifier.
7. Train the classifier.
8. Predict loan approval for test data.
9. Calculate the accuracy.
10. Display the prediction results.

Output:



```
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.naive_bayes import GaussianNB
4 from sklearn.preprocessing import LabelEncoder
5 from sklearn.metrics import accuracy_score
6 data = pd.DataFrame({'Age': [25, 30, 35, 40, 45, 28, 32, 50, 38, 27], 'Income': [25000, 40000, 50000, 70000, 80000, 30000, 45000, 90000, 60000, 28000],
7   'CreditScore': [600, 650, 700, 750, 780, 620, 690, 800, 720, 610], 'LoanApproved': ['No', 'No', 'Yes', 'Yes', 'Yes', 'No', 'Yes', 'Yes', 'Yes', 'No']})
8 encoder = LabelEncoder()
9 data['LoanApproved'] = encoder.fit_transform(data['LoanApproved'])
10 X = data[['Age', 'Income', 'CreditScore']]
11 y = data['LoanApproved']
12 X_train, X_test, y_train, y_test = train_test_split(
13     X, y, test_size=0.3, random_state=42)
14 model = GaussianNB()
15 model.fit(X_train, y_train)
16 y_pred = model.predict(X_test)
17 print("Predicted:", y_pred)
18 print("Actual:", y_test.values)
19 print("Accuracy:", accuracy_score(y_test, y_pred))
20
```

Run exp 19

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/exp 19.py
Predicted: [1 1 0]
Actual: [1 0 0]
Accuracy: 0.6666666666666666
Process finished with exit code 0
```

Result: The Naive Bayes classifier successfully predicted whether customers would be approved for a bank loan.